

Activity Guide — middle / module-15-project-implementation

IKS Curriculum

Table of Contents

Activity 1 — Mid-Project Critique Circle

Module: Project Implementation — Capstone · **Band:** Middle · **Duration:** 30 min

Purpose

Reinforce the day-by-day concepts of Project Implementation — Capstone through a hands-on activity. Aligned with NEP 2020 §4.23 (experiential learning).

Materials

student project plans, sticky notes

What students do

Pairs swap project plans, read in 5 minutes, then offer one strength (one sticky note) and one concern (another sticky note). Pairs rotate twice. Refine plan based on feedback.

Step-by-step

1. **Setup (5 min)** — gather materials; explain the task.
2. **Working phase (20 min)** — students work individually or in pairs.
3. **Share (5 min)** — selected students share their results with the class.

Differentiation

- **Extension:** ask advanced students to do an additional iteration or analysis.
- **Support:** pair students who need scaffolds; offer partial templates.

Assessment

This activity is **formative**. The teacher observes participation, accuracy, and reflection quality. No grade is assigned.

Safety

- General safety briefing before any materials are handled.

- If herbs, plants, or food are involved: kitchen-amount only, no ingestion of unknown preparations.
- Outdoor activities: stay in the designated area; supervised by the teacher.

Activity 01 — Source Hunt

Module: Project Implementation · **Band:** Middle · **Day:** 3 · **Time:** 25 min (within Day 3's 45 min)

Type

Individual research, with cluster-based peer help

Goal

Each student locates at least two verifiable sources for their project and records each on a citation slip.

Materials per cluster

- Printed `sources.md` of the anchor module
- Computer or tablet (shared) to open `.rag-cache/` JSON files
- Citation slips (pre-printed; see template in `days/day-03-research-1.md`)
- Pen
- Sticky tabs for marking pages

Set-up (teacher, before class)

- Group desks into clusters by anchor module. Label each cluster.
- Put one printed `sources.md` (or its PDF on a shared laptop) per cluster.
- Pre-stage the cache files on the cluster laptop so they open in one click.

Instructions (student-facing — printable handout)

Source Hunt — individual work, ask your cluster-mates if stuck.

1. Open your anchor module's `sources.md`. Skim for entries that match your project's question.
2. Open the `.rag-cache/` folder for that module. Each file is named for the day it was used; open the one most relevant to your topic.
3. For each candidate source, read the chunk in full. Don't just take the one verse — read the surrounding lines.
4. **Fill in a citation slip** for any source you'll actually use.
5. Stop when you have **at least two** completed slips. Three is better.
6. **If you get stuck:** ask another student in your cluster first. Then ask the teacher.

Two-slip minimum

Slip #1 must come from **category 1** (primary text) or **category 2** (secondary reference).

Slip #2 can be from any of the three categories: primary text, secondary reference, or primary observation.

(If you go to a third slip, mix categories — triangulation makes your project stronger.)

Worked example — Project 4 (herb deck)

Slip #1 - Source identifier: Caraka Saṃhitā Sūtrasthāna 1.41 (cached in module-10-herbs-aushadhi/.rag-cache/day-02-aushadhi-intro.json#chunks[1])
- What it says: Plants used as medicine are not just collections of parts — each part (root, leaf, bark, fruit, seed) has its own action and is collected at a particular time of year.
- How this relates to my project: I'll add a "part used" line to each herb card. This is why.
- Category: primary text.

Slip #2 - Source identifier: Pradip Krishen, *Trees of Delhi* (Penguin Books, 2006), entry on *Azadirachta indica* (neem).
- What it says: Neem grows ubiquitously in Delhi and parts of north India; the leaves are bitter, the bark is astringent, the fruit is sweet.
- How this relates to my project: I'll cite this for the neem card; same author/book may have several of my 10 herbs.
- Category: secondary reference.

Discussion prompts (last 5 min, whole class)

- Which source surprised you?
- Did anyone find a source that they thought would be irrelevant but turned out to be useful?
- Did anyone find that their anchor module's cache doesn't have what they need — and what did you do about it?

Looks-like / sounds-like

	Looks like	Sounds like
Going well	Student is reading a cache file with a pen in hand, taking notes in their own words	"This one's about the same thing as my project — let me write it down"
Needs intervention	Student staring blankly at the cache file, scrolling without reading; or copying full chunks word-for-word	"I'll just paste this whole thing in" — intervene: "what does it say in your own words?"

Safety / sensitivity

- No safety concerns specific to this activity.
- If a student finds a source that challenges their starting assumption (e.g. their fermented-foods project anchor turns out to disagree with what their grandmother taught them), honour both — note the disagreement, don't force resolution.

Differentiation

- **Tier up:** After two slips, identify one *opposing* source — something that complicates or qualifies the first two. This is good intellectual practice and earns a bump on the rubric’s source-use criterion.
- **Tier down:** Focus on one excellent slip rather than two mediocre ones. Quality beats quantity. Teacher signs off as completion.

Clean-up

- All citation slips go into the project journal.
- Cache files stay on the shared laptop for Day 4.

Assessment note (teacher)

Mark each slip on a quick checklist: - ✓ Identifier present? - ✓ “Where I found it” specific (file path, not just module name)? - ✓ Paraphrase in student’s own words (not copied verbatim)? - ✓ “How this relates” linked to the project’s driving question?

A slip with all four ticks is a Meets (3) on the rubric. Slips with two or fewer ticks need re-work tomorrow.

Activity 02 — Paired Critique Protocol

Module: Project Implementation · **Band:** Middle · **Day:** 5 · **Time:** 25 min (within Day 5’s 45 min)

Type

Paired structured critique, both directions

Goal

Each student gives and receives one full round of “I notice / I wonder” feedback on their plan, sources, and sketch — and walks out with three specific changes they’ll make on Days 6–7.

Materials per pair

- Both students’ plan + citation slips + sketch
- Two printed **Critique Protocol Sheets** (one per direction; template in `days/day-05-critique.md`)
- Pen each
- Timer (phone)

Set-up (teacher, before class)

- **Pair students for variety** — pair across project types if possible (Project 1 with Project 6, Project 4 with Project 7, etc.). Pairs *within* the same project type tend to mirror; cross-project pairs surface real differences.

- 14 pairs maximum for a class of 28. If odd, one trio (more challenging — see notes).
- Print the protocol sheets in advance.

Instructions (student-facing — printable handout)

Paired Critique Protocol — work in your assigned pair.

One direction at a time. Set a timer.

Round 1 — Student A presents, Student B critiques.

Step	Time	What happens
1	3 min	A presents. Walks B through the plan, sources, sketch. B listens. B does NOT interrupt.
2	2 min	B asks clarifying questions only. No suggestions, no judgements. <i>Just</i> questions.
3	4 min	B gives “I notice / I wonder.” Three “I notice...” + three “I wonder...”, written on the protocol sheet.
4	3 min	A responds. What I’ll keep / change / think about. A writes on their own protocol sheet.

Round 2 — swap. Same structure.

End of activity (3 min): joint reflection. *What does this comparison reveal about both your projects?*

“I notice...” — three rules

- **Specific.** “Your sketch is nice” doesn’t count. “Your sketch has the title in the centre, which pulls my eye away from the data on the left” counts.
- **Neutral.** Describe what you see. Don’t judge yet.
- **About the work, not the person.** “Your colour scheme is grey-on-grey” not “you don’t like colour.”

“I wonder...” — three rules

- **A question or a suggestion.** “I wonder what would happen if you swapped panels 2 and 3?” or “I wonder if the title block could be smaller.”
- **Helpful, not prescriptive.** It’s a suggestion, not an order.
- **Tied to a specific element.** Not “I wonder if it’s good overall” — point at one thing.

Discussion prompts (last 3 min, joint reflection)

- What’s one strength both your projects share?
- What’s one weakness both your projects share?
- Which project is more ambitious right now? Should it be?
- If you had to swap projects with each other for Days 6–7, what would you find hardest?

Looks-like / sounds-like

	Looks like	Sounds like
Going well	Both partners pointing at the sketch, asking real questions, writing on the sheet	“What’s the dotted line for?” / “I notice your sources are all from one cache — I wonder if a second cache would balance it”
Needs intervention	One partner doing all the talking and praising; the other silent or defensive	Silence; “It’s good.” “Thanks.” — intervene: “B, give A one specific ‘I notice’”

Safety / sensitivity

- **Emotional safety.** A critique can sting when a student has worked hard. The “I notice / I wonder” structure is designed to take the sting out — it puts the work, not the person, at the centre. Re-direct any pair that drifts into personal language.
- If a critic gives feedback that the presenter finds genuinely unfair, the presenter can write “disagree” on the sheet and we’ll discuss in 1:1 tomorrow morning. No public arguing in the room.

Differentiation

- **Tier up:** After both rounds, write a paragraph in your journal about what you noticed about your *own* assumptions when you heard your partner’s critique. Did anything they said reveal something you took for granted?
- **Tier down:** Two “I notice” and two “I wonder” per round, not three. Quality of two specific observations beats six vague ones.

Trio (if class is odd)

If you have one trio rather than 14 pairs:

- Three rounds (A → B, B → C, C → A) at 10 minutes each. Fits in 30 minutes total instead of 25 — borrow time from the Day 5 wrap-up.
- Each student presents once and critiques once.
- The trio reflection at the end is richer — three perspectives on each project.

Clean-up

- Both completed protocol sheets stay with the presenter — pin them into the project journal.
- The “three changes” written by each student is the input to Day 6’s build session — keep it visible.

Assessment note (teacher)

This activity isn't directly graded. But two markers feed the rubric's *process documentation* score:

- Did the critique sheets get filled in fully (both presenter response and critic notes)?
- Did the student's Day 6 journal entry refer back to a specific critique point?

If both are present — solid 3 on process documentation. If neither — drop.

Activity 03 — Rehearsal Run-Through

Module: Project Implementation · **Band:** Middle · **Day:** 9 · **Time:** 27 min (within Day 9's 45 min)

Type

Paired full-share rehearsal with structured feedback

Goal

Each student delivers a complete 4-minute share + 1-minute Q&A to a single peer and receives structured written feedback before Day 10.

Materials per pair

- Each student: artefact + documentation + any notes / cue cards
- Timer (phone)
- Two printed **Rehearsal Feedback Sheets** (template in `days/day-09-rehearse.md`), one for each direction
- Pen each

Set-up (teacher, before class)

- Same pairs as Day 5 (continuity of feedback rapport).
- Set up the room so pairs can present somewhat in private — not 14 simultaneous performances in one open space (the noise floor wrecks the rehearsal). Use corners, hallway corners, or staggered start times if needed.
- A teacher demonstration (90 seconds, optional) at the very start, showing what a tight 4-minute share looks like — pick one project type and model it.

Instructions (student-facing — printable handout)

Rehearsal Run-Through — work in your assigned pair.

Two directions, ~12 minutes each. The structure for each direction:

Step	Time	What happens
1	4 min	Presenter delivers the share, end to end. Listener

Step	Time	What happens
		does NOT interrupt — even if confused.
2	1 min	Listener asks ONE question. Pick the one that genuinely puzzled you. The presenter answers in 30 seconds.
3	5 min	Listener fills in the Feedback Sheet. Two specifics on “what landed” + two on “what was unclear” + one refinement.
4	2 min	Presenter reads the sheet silently. No defending. Writes their three refinements for tomorrow on their own copy.

Then **swap**.

End of activity (3 min): both presenters do a final silent revision of their three refinements. Pin the feedback sheet into the project journal.

Timing standards

- **3:45–4:15** = on time (within range)
- **3:30–3:45 or 4:15–4:30** = needs one cut or one expansion
- **Outside that range** = re-plan tonight; the share itself is timed strictly

Q&A practice

The single question your listener asks is your practice Q&A. Two acceptable responses:

1. **You know the answer** — give it concisely in 30 seconds.
2. **You don’t know the answer** — say so honestly: *“I don’t know, but here’s what I’d check...”* This is a legitimate answer and the rubric does not penalise it.

What you should NOT do: - Pretend to know and bluff. - Get defensive. - Take more than 30 seconds (Q&A tomorrow is 60 seconds *total*).

Discussion prompts (built into Step 4)

- Which part of your share felt most natural?
- Which part felt rushed?
- Did you forget anything that’s on your documentation?
- What’s one word you’d change in your opening sentence?

Looks-like / sounds-like

	Looks like	Sounds like
Going well	Presenter is standing, holding the artefact at the right moment; listener is timing and writing	“30 seconds left, you’re on Sources” / “I had to write fast to keep up — that’s a good thing”

	Looks like	Sounds like
Needs intervention	Presenter reading off notes word-for-word with no eye contact; listener checking their phone	Monotone reading; intervene: “Look up. Tell me, don’t read.”

Safety / sensitivity

- Some students will be very nervous. Permit them to do an “untimed” first run, then a “timed” second run. The rubric only sees the timed one.
- A student who genuinely cannot present live can submit a video by Day 10 morning instead. Same rubric.

Differentiation

- **Tier up:** Deliver your second rehearsal *without notes*. The first rehearsal allows notes; the second must be from memory. If the second feels worse than the first, return to notes for tomorrow — the memory attempt was practice, not a downgrade.
- **Tier down:** Use cue cards tomorrow. Five cards (one per section). Large handwriting. Hold them low and look up between cards.

Clean-up

- Both feedback sheets pinned into the project journal.
- Each student writes their three refinements in their journal as a numbered list.
- Pack the artefact carefully tonight — it travels home and back tomorrow.

Assessment note (teacher)

The rehearsal itself is not graded. But the teacher can do a quick “drive-by” check during each pair’s first round and note:

- Was the share within range (3:45–4:15)?
- Did the structure follow Question → Process → Sources → Artefact → Reflection?
- Did the reflection include a specific “what didn’t work”?

These three checks tell you who will land tomorrow’s share well, and who needs a 5-minute morning huddle before Day 10.

Project Brief

Project Brief — Module 15 Capstone (Middle)

Work sessions: Days 1–9 (in class) · **Share + summative:** Day 10 · **Total marks:** 40

The project

Each student (or pair) picks **one** project from the menu (or proposes their own under Project 10) and runs it through the **six phases**: scope → research → design → build → document → share.

The artefact format depends on the project. The discipline is the same across all projects.

The menu

#	Project	Anchor module	Format suggestion
1	Fermented-foods comparative study (the safer adaptation of the Pañcagavya theme — see safety note in README)	Module 3 (Ayurveda) + Module 4 (Doṣa)	Poster with experimental setup + tasting-notes table
2	Mandala / prthvī geometry design	Module 5 + Module 2	Hand-drawn mandala on A3 with colour key + construction notes
3	Tithi + saṁkrānti + festival calendar	Module 6 + Module 12	A printed/drawn one-month or three-month calendar
4	Herb identification card-deck	Module 10	10 postcard-size cards, bound or in a small envelope
5	Magic-square pattern art	Module 14	A3 framed grid with the verified magic square + art overlay
6	Yoga sequence design	Module 8	Storyboard sheet + practice card for the target age group
7	School sustainability audit	Module 13	1–2 page audit report with measured data + 3 recommendations
8	Doṣa daily-rhythm self-study (observation only, NOT diagnosis)	Module 4	7-day journal in a structured table + 1-page reflection
9	Multiplication -method	Modules 7, 9, 11	Data sheet + 1-page write-up of mean time and error rate per method

#	Project	Anchor module	Format suggestion
	comparison (timed)		
10	Open-design project — student-proposed, teacher-approved	Any	Format to be approved with the plan

The six-phase arc

Phase	Days	What you do
Scope	1–2	Pick the project, write the one-page plan
Research	3–4	Find at least 2 verifiable sources; first design sketch
Critique	5	Paired structured feedback (“I notice / I wonder”)
Build	6–7	Make the artefact (80% complete by end of Day 7)
Document	8	Write the five-section process documentation
Share	9–10	Rehearse + deliver 4-minute share with Q&A

What every project must include

1. **A driving question** — written in one sentence on the Day 2 plan.
2. **At least two verifiable sources** with specific identifiers (file path, page number, or canto.chapter.verse).
3. **A finished artefact** in the chosen format. Finished beats impressive.
4. **A five-section documentation** — question, process, sources, artefact, reflection.
5. **A 4-minute presentation** with a 1-minute Q&A, delivered on Day 10.
6. **An honest reflection** that names something specific that didn’t work.

What success looks like

A strong project will:

- Apply the chosen IKS framework correctly and show awareness of its limits.
- Cite at least two verifiable sources, and ideally one that *complicates* the central claim.
- Be visibly finished, with the artefact answering the driving question.
- Show the student’s own thinking — not just a summary of someone else’s work.
- Be communicated clearly in the share — including a specific moment of “what didn’t work.”

Working in pairs (or individually)

- **Pairs are encouraged** for projects that benefit from a build partner (Projects 4, 5, 6) and for projects involving primary observation (Projects 3, 7).
- **Solo is allowed** for any project; required for Projects 8 (doṣa self-study) and 10 (open design, by default).
- Groups of 3 are acceptable for Projects 7 (sustainability audit) only, with each member having a distinct role.

Citation rules

- At minimum **two** verifiable sources must appear in your documentation.
- Sources must have specific identifiers (see the “what does NOT count” list in Day 3’s lesson).
- An invented citation goes to 0 on Use of Sources AND triggers a re-do offer with a deadline.

Schedule

Day	What you do	Submit at end of class
1	Tour the menu, shortlist two options	Sticky note on the board
2	Write the one-page plan	Plan submitted
3	First sources	Two citation slips
4	More sources + first sketch	Sketch + third slip
5	Paired critique	Two protocol sheets pinned in journal
6	Build session 1	Photographed progress
7	Build session 2	Photographed progress; artefact 80% done
8	Documentation	Five-section write-up
9	Rehearsal	Feedback sheet + 3 refinements written
10	Share + summative quiz + reflection	Final artefact + documentation + journal

Self-reflection (submitted with project on Day 10)

In 4–6 sentences:

1. What did you set out to find / make, and what did you actually end up with?
2. What is the most surprising thing you discovered while working on this?
3. What specifically didn’t work — and what would you change with three more days?

Pañcagavya safety note (READ before choosing Project 1)

The traditional Pañcagavya preparation includes cow urine and cow dung. **This curriculum does NOT permit students to handle, prepare, taste, or ingest pañcagavya.** The school-safe adaptation is the fermented-foods comparative study (Project 1 above). A library-only research

project on pañcagavya is permitted with explicit written parent consent and teacher approval — no handling, balanced sources, written report only. See `teacher-notes.md` for the full protocol.

Rubric

See `rubric.md`. Read it before Day 8.

Exemplars

Strong examples from past cohorts (anonymised, with permission) will be added to `exports/pdf/middle-module-15-project-implementation/exemplar-*.pdf` after the first cohort presents.